

Properties of Binary Relation

~ Consider Set A,
Binary Relation on Set A is
 $R \subseteq A \times A$

Ex $A = \{1, 2\}$
 $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$
 $R = \{(1, 1), (1, 2)\}$

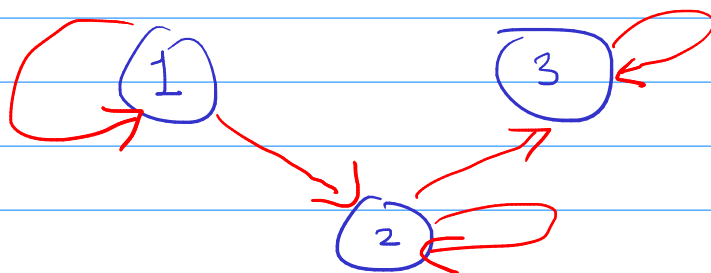
1. Reflexivity

~ R is defined on A. for every $a \in A$
If $(a, a) \in R$, then R is Reflexive.
 $A = \{1, 2, 3\}$

$$R = \{(1, 1), (1, 2), (2, 2), (2, 3), (3, 3)\} - \checkmark$$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} \textcircled{1} & & \\ & \textcircled{1} & \\ & & \textcircled{1} \end{bmatrix} \end{matrix}$$

"Each Cell on diagonal Contains 1"



"Each Vertex Contains Self Loop"

Ex. $A = \{1, 2, 3, 4\}$, $R = \{(a, b) \mid a > b\}$
~ Is R Reflexive? — NO

$$R = \{(2, 1), (3, 1), (3, 2), (4, 1), (4, 3), (4, 2)\}$$

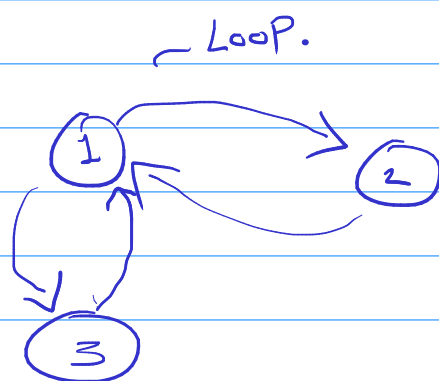
~ If $R = \{(a, b) \mid a > b\}$ — YES

2. Symmetry:

- If R is Binary Relⁿ on A , then for each pair $(a, b) \in R$, if $(b, a) \in R$ then R is Symmetric Relation.

$$R = \{(1, 2), (2, 1)\} - \checkmark$$

$$R = \begin{matrix} & \begin{matrix} 1 & 2 \end{matrix} \\ \begin{matrix} 1 \\ 2 \end{matrix} & \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \end{matrix}$$



~ $\mathcal{U} = \{(a, a), (b, b)\}$ — Both
Symmetric & Reflexive.

$X = A \times A$ — Symmetric — $\checkmark$
Reflexive — $\checkmark$

$$A = \{\overline{1}, \overline{2}\}$$

$$= \{(\underline{1}, \underline{1}), (\underline{1}, \underline{2}), (\underline{2}, \underline{1}), (\underline{2}, \underline{2})\}$$

~ Anti-Symmetric Relation.

* R is Relation on A . If $(a,b) \in R$ and $(b,a) \in R$ Implies $a=b$ then R is anti Symmetric Relation.

~ $(a,b) \in R$ then (b,a) Should not $\in R$.
 (b,a) Should $\in R$ only if $a=b$. (means it is allowing reflexive Pairs)

~ $A = \{1, 2, 3\}$
 $R = \{(1,1), (1,2), (1,3), (3,3), (2,1), (2,2)\}$
— Not anti-Symmetric

$R = \{(1,1), (1,2), (1,3), (3,3), (2,2)\}$
— Anti Symmetric

Ex $A = \{a, b, c\}$ Symmetric
 $S = \{(a,a), (b,b)\}$ Anti Symmetric
Both — ✓

$N = \{(a,b), (a,c), (c,a)\}$

— Symmetric — X
— Anti " — X
— Both

* IS every Reflexive anti Symmetric? NO
 $A = \{a, b, c\}$
 $R = \{(a, a), (b, b), (c, c), (a, b), (b, a)\}$

(3) * Transitivity

~ R is Ref^n on A . If $(a, b) \in R$ and $(b, c) \in R$, Implies $(a, c) \in R$ then R is Transitive Relation.

$$A = \{a, b, c\}$$

$$X = \{(\underline{a}, a), (\underline{a}, b), (a, c), (b, c)\}$$

Ex. Let T be a binary Relation on A Such that (a, b) is in T if and only if a is father of b

- IS T Transitive?

~ Assume T is Transitive

$$T = \left\{ \underset{\substack{F \quad S}}{(a, b)}, \underset{\substack{F \quad S}}{(b, c)}, \underset{\substack{F \quad S}}{(a, c)} \right\}$$

- $(c, a) \notin R$, So T is non Transitive.

* Closure of the Relation

- ↳ Reflexive Closure
- ↳ Symmetric Closure
- ↳ Transitive Closure

* Reflexive Closure: of Relation R is
"Smallest Reflexive Relation that
Contains R " (R')

Ex. $R = \{(1,1), (2,2), (2,3)\}$

$$\begin{aligned} R_1(R) &= R \cup \{(a,a) \mid a \in A\} \\ &= \{(1,1), (2,2), (2,3)\} \cup \{(1,1), (2,2), (3,3)\} \\ R_2(R) &= \{(1,1), (2,2), (2,3), (3,3)\} \end{aligned}$$

Ex. $A = \{0,1,2,3\}$
 $R = \{(0,1), (1,2), (0,0), (2,2), (3,2), (1,3)\}$
 $\quad\quad\quad (1,1), (3,3)\}$

* Symmetric Closure of Relation R is Smallest Symmetric Relation that contains R .

$$- S(R) = R \cup \{ (b, a) \mid (a, b) \in R \}$$

$$A = \{0, 1, 2\}$$

$$R = \{ (0, 1), (1, 2), (2, 1) \}$$

$$\begin{aligned} S(R) &= \{ (0, 1), (1, 2), (2, 1) \} \cup \{ (1, 0), (2, 1), (1, 2) \} \\ &= \{ (0, 1), (1, 2), (2, 1), (1, 0) \} \end{aligned}$$